

AY: 2025-2026

Course: PPOM 1

Class of 2029

Week 14, Lecture 134

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DPR Lecture: History and Physical Exam of the Hip and Knee

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Session Objectives

- Demonstrate the components of the history taking and physical examination of the Hip and Knee
- Evaluate subjective and objective findings relating to the examination of the hip and knee
- Distinguish Normal vs. Abnormal examination findings
- Recall the appropriate way to document common normal and abnormal findings of the history and physical Examination

RED WOLVES and BEARS

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The Health History

- Common or Concerning Symptoms
 - Joint pain: articular or extra-articular, acute or chronic, inflammatory or noninflammatory, localized or diffuse, monoarticular or polyarticular
 - Joint pain: associated constitutional symptoms and systemic manifestations from other organ systems
 - Fever, chills, rash, weight loss, weakness
 - Neck pain
 - Low back pain

Tips for Assessing Joint Pain

- “Point to the pain”
- Determine whether the pain is
 - articular or extra-articular
 - acute or chronic
 - inflammatory or noninflammatory
 - localized (monoarticular) or diffuse (polyarticular)

History Taking- Joint Pain

- Is the pain monoarticular or polyarticular?
- What is the pattern of involvement?
 - Migrating from joint to joint
 - Spreading from one joint to multiple joints
- Is the involvement symmetric?

History Taking – Joint Pain

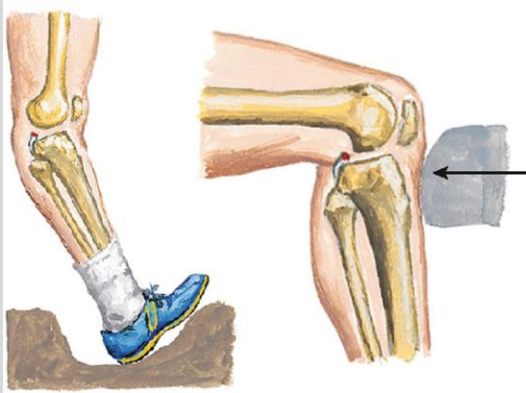
- Articular
 - True pain from the joint
- Extra-Articular
 - Involves bones, muscles, and tissues around the joint
 - Occurs in inflammation of bursae, tendon, or tendon sheaths
 - Sprains- stretching or tearing of ligaments

Tips for Assessing Joint Pain

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- Clarify and record the Mechanism of injury



PCL Injury

Usual causes include hyperextension injury, as occurs from stepping into hole, and direct blow to flexed knee



Sprains

Usual cause is forceful impact on posterolateral aspect of knee with foot anchored, producing valgus stress on knee joint



ACL Injury

Usual cause is twisting of hyperextended knee, as in landing after basketball jump shot

History Taking – Joint Pain

- Inflammation
 - 4 Cardinal features of inflammation
 - Pain
 - Swelling
 - Warmth
 - Redness
 - Causes
 - Infectious, Crystal-induced, Immune-related, Reactive, or Idiopathic
- Non-inflammation
 - Trauma, Repetitive use, degenerative changes, or fibromyalgia

History Taking – Joint Pain

- Assessed decreased or limited movement
- Timing
 - Acute vs. Chronic
 - What aggravates or relieves the pain
- Systemic Features
 - Fever? Chills? Rash? Fatigue? Anorexia? Weight loss? Weakness?

Common Review of Systems

- Constitutional: Fever, chills, or fatigue
- Derm: Rash or sores
- HEENT: Head injury, blurry vision, ear pain or sinus pain
- Neck: Neck pain
- Respiratory: SOB or difficulty breathing
- CV: Chest pain or palpitations
- GI: N/V/D/C; Abdominal pain

Common Review of Systems

- Peripheral Vascular: No varicose veins
- MSK: Any other joint pain, muscle pain/weakness
- Neuro: Numbness / Tingling / Weakness
- Endocrine: Excessive thirst / hunger / urination
- GU: F / U / D; Dysuria; Discharge
- Psychiatric: Depression or anxiety

The Hip and Knee- Thinking about the Diagnosis

- Age
- Pain
 - Onset
 - Location
 - Occurrence
- Snapping
- Assisted Ambulation
 - What are you using to help you walk around?
 - How often do you rely on assisted ambulation?
- Activity Tolerance
 - What Activities are you limited to
 - Change in Distance

The Hip and Knee- Thinking about the Diagnosis

- Trauma
 - Has there been any associated trauma?
 - Including MVA, Falls, “Hard hits”
- Activity/Work/Sports
 - What kind of work do you do?
 - How active are you daily?
- Neurologic Symptoms
 - Any associated numbness or tingling?
- History of Multiple Joints pain
 - Any other joint pain?

The Hip

- Ball and Socket joint
- Unique set of factors contributing to its stability and function
 - Muscular attachments
 - Cartilaginous and ligamentous structures
 - Alteration of the relationship between the femur and acetabular fossa and result in undue wear and disruption of the cartilaginous surfaces

The Hip

- Acetabular-femoral joint is encased by a fibrous capsule forming the central joint cavity
- It originates at the acetabular rim and inserts on the intertrochanteric line of the femur and its neck posteriorly
- The Capsule is lined by a synovial layer that secretes the viscous synovial fluid

The Hip

- Muscular support of the hip is described by function
- Primary movements → Flexion and Extension
- Iliopsoas serves as the primary flexor
- Gluteus Maximus serves as the primary extensor

The Knee

- Primarily functions as a hinge joint with the articulation b/t the femur and the tibia serving as the main load bearing
- Distal femur → bone widens to form Medial and lateral condyles
- Distal femur and proximal tibial articular surfaces are covered with hyaline cartilage
- Patella → large sesamoid bone that functions as the insertion of the quadriceps tendon and the origin of the infrapatellar ligament
 - Anterior stabilization and
- Fibula → Allow rotation of the lower leg without sacrificing stability
 - The proximal end of the fibular head articulates medially with the lateral portion of the tibial plateau

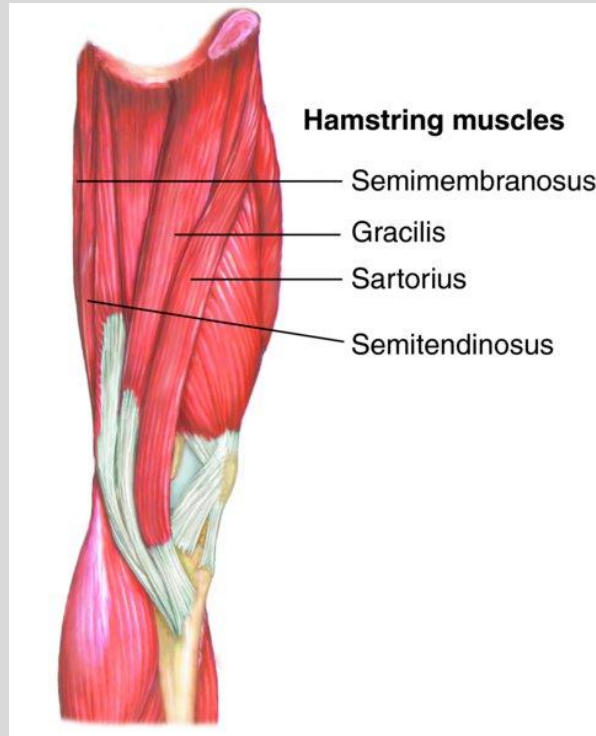
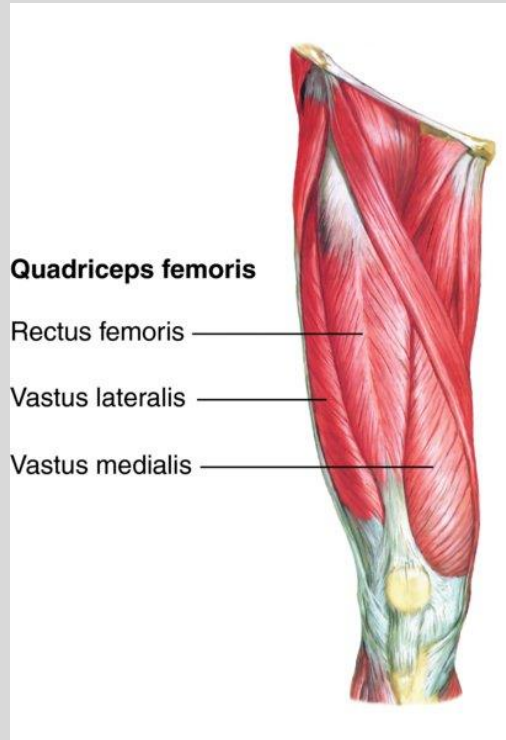
The Knee - Muscles

- Extensors
 - Quadriceps
 - Vastus medialis
 - Vastus intermedius
 - Vastus lateralis
 - Rectus femoris
 - Tensor fascia latae
- Flexors
 - Hamstring muscles (also cross the hip and act as extensors of the hip)
 - Semitendinosus
 - Gracilis
 - Long head of the biceps femoris
 - Gastrocnemius and plantaris are also knee flexors

The Knee - Muscles

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Source: Bates' Guide to the Physical Examination and History Taking; 12th Ed. – Chapter 16

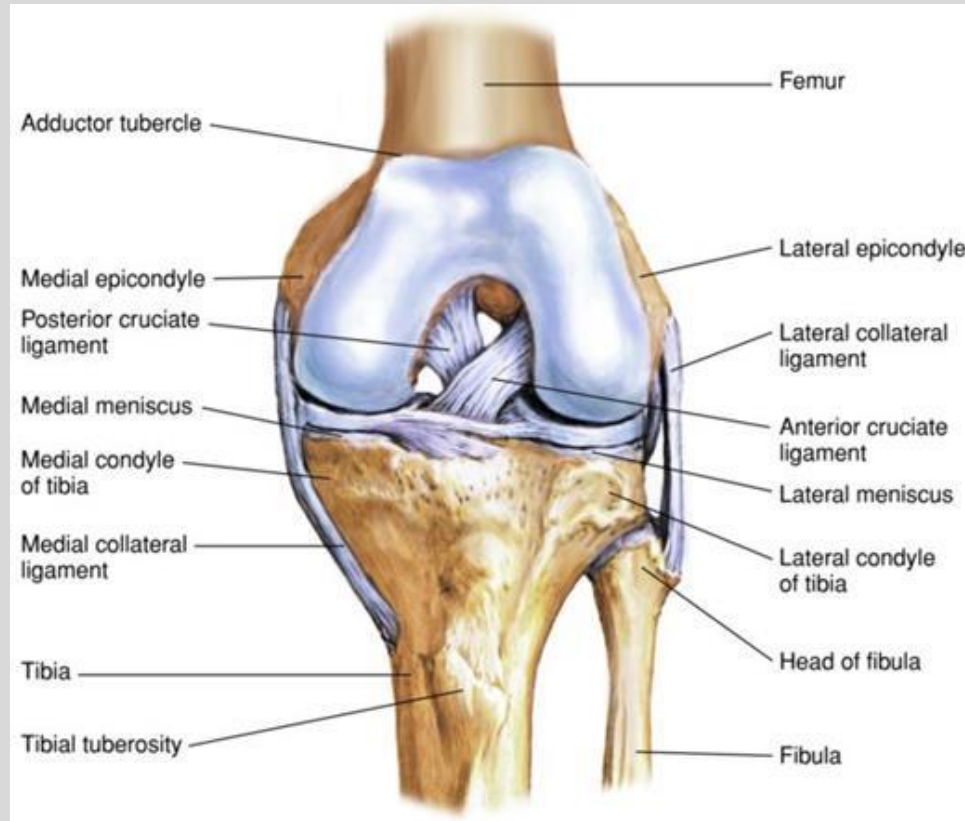
The Knee - Ligaments

- Anterior Cruciate Ligament (ACL)
 - Crosses anterior medial tibia to the lateral femoral condyle (prevents tibia from sliding forward on the femur)
- Posterior Cruciate Ligament (PCL)
 - Crosses posterior tibia to the medial femoral condyle (prevents tibia from sliding backward on the femur)
- Medial Collateral Ligament (MCL)
 - Connects medial femoral epicondyle to the medial condyle of the tibia
 - Also attaches to the medial meniscus
- Lateral Collateral Ligament (LCL)
 - Connects lateral femoral epicondyle and the head of the fibula

The Knee - Ligaments

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Source: Bates' Guide to the Physical Examination and History Taking; 12th Ed. – Chapter 16

The Knee - Menisci

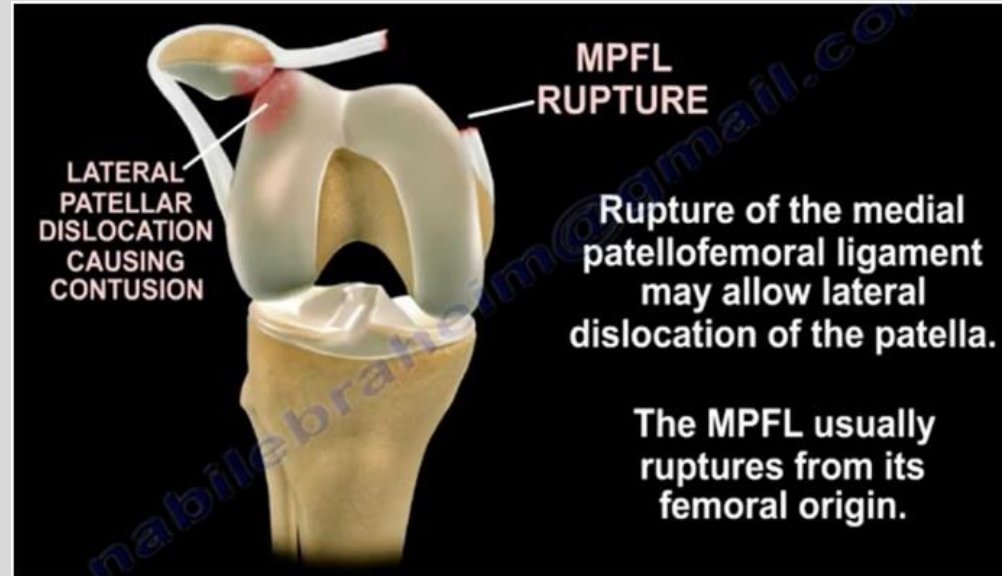
- “C” shaped cartilaginous structures that attaches to the tibial plateau by the menisco-tibial ligaments
- Primarily help stabilize the femoral-tibial articulation
- Help with weight transfer, knee proprioception and nutritional functions for the articular cartilage

The Knee – Medial Patellofemoral Ligament

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- Ligamentous stabilizer of the patella
- Essential role in resisting natural lateral migration of the patella
- Often ruptured in patellar dislocation
- When torn, you can see lateral patellar instability along with lateral migration of the patella



Krebs C, Tranovich M, Andrews K, Ebraheim N. The medial patellofemoral ligament: Review of the literature. *J Orthop*. 2018;15(2):596-599. Published 2018 May 7. doi:10.1016/j.jor.2018.05.004. Link: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5990246/>

Physical Examination- Hip and Knee

- Should be systematic
 - Examine gait
 - Inspection
 - Palpation
 - Range of Motion
 - Active
 - Passive
 - Strength testing
 - Additional testing

Physical Examination- Hip and Knee

- Assessing the 4 signs of Inflammation
 - Swelling
 - Warmth
 - Redness
 - Pain or tenderness
 - Diffuse
 - Focal

Lower Extremity Exam- Examine Gait

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- Gait
 - Walk, turn and come back
 - Heel-to-toe tandem walking
 - Shallow bilateral knee bend squat
- Note any signs of imbalance, type of gait, postural asymmetry



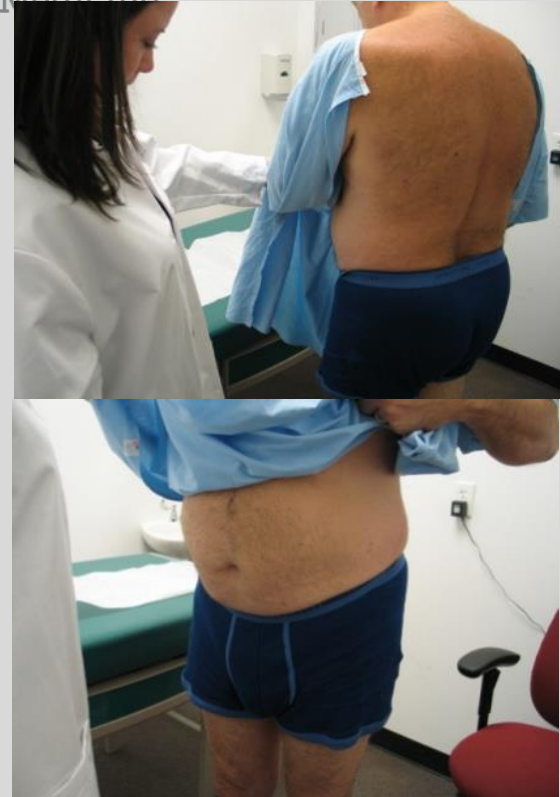
Hip Exam- Inspection

- Bilateral Inspection
- Clearly Inspect the hips
- “I’m inspecting your hips”
- Inspects anterior, posterior and lateral
- Examiner inspect hips for abnormalities, asymmetry, swelling, atrophy of the muscle



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Knee Exam- Inspection

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- Bilateral Inspection
- Clearly Inspect the knees
- “I’m inspecting your knees”
- Inspects anterior, posterior and lateral
- Examiner inspect knees for abnormalities, asymmetry, swelling, atrophy of the muscle



Knee Exam- Inspection

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- Bilateral Inspection
- Clearly Inspect the knees
- “I’m inspecting your knees”
- Inspects anterior, posterior and lateral
- Examiner inspect knees for abnormalities, asymmetry, swelling, atrophy of the muscle

Hip Exam- Palpation

- Bilateral exam
- Use finger pads for palpation
- Patient in lateral recumbent position
- Inguinal region – check for lymph nodes or hernia
- Note: Ligament or tendon tenderness after injury is suspicious for tear



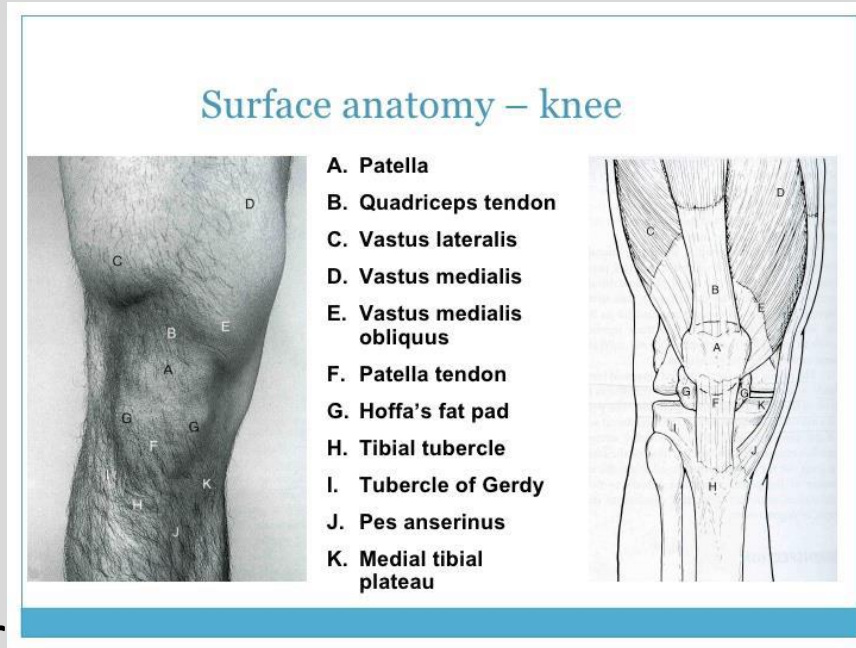
- Landmarks
 - Greater Trochanter
 - Anterior Superior Iliac spine (ASIS)
 - Anterior Inferior Iliac spine (AIIS)
 - Posterior superior iliac spine (PSIS)
 - Sacroiliac joint
 - Pubic tubercle
 - Iliac crest
 - Ischial tuberosity

Knee Exam- Palpation

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- Bilateral palpation
- Use finger pads for palpation
- Patient in seated position
- Palpate for tenderness



- Landmarks
 1. Patella
 2. Tibial tubercle
 3. Medial / lateral epicondyle of femur
 4. Medial / lateral epicondyle of tibia
 5. Medial / lateral knee joint line
 6. Medial / lateral collateral ligament.
 7. Patellar ligament
 8. Iliotibial band
 9. Fibular head at its styloid process

Hip Exam- Passive Range of Motion Testing

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- Bilateral exam
- Gown covering patient's thighs and above



Flexion



Extension



External Rotation



ABDuction



ADDuction



Internal Rotation

Source: Bates' Guide to
the Physical Examination
and History Taking; 12th
Ed. – Chapter 16

Hip Exam- Active Range of Motion Testing

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- Bilateral exam
- Gown covering patient's thighs and above



Flexion



Extension



External Rotation



ABDuction



ADDuction



Internal Rotation

Source: Bates' Guide to
the Physical Examination
and History Taking; 12th
Ed. – Chapter 16

Hip Evaluation

- Hip Range of Motion:
 - Flexion $120^{\circ} - 130^{\circ}$
 - Extension $10^{\circ} - 20^{\circ}$
 - Abduction $45^{\circ} - 60^{\circ}$
 - Adduction $30^{\circ} - 40^{\circ}$
 - Internal Rotation $35^{\circ} - 40^{\circ}$
 - External Rotation $40^{\circ} - 50^{\circ}$

Sources:

1. Value reproduced from PowerPoint Presented to Class of 2022- *Foundations of Osteopathic Medicine*- OMM Lab #11 – “Osteopathic Evaluation of the Hip” presented November 14th and 15th, 2018

Flexion 120° – 130°

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- Note: Flexion of the hip with the knee extended will test the flexibility of the hamstrings

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Extension $10^{\circ} - 20^{\circ}$

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ABDuction 45° – 60°

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ADDuction 30° – 40°

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External Rotation 40° – 50°

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- Note: Flex hip to 90 degrees to test
- May also be tested prone or seated

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Internal Rotation 35° – 40°

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- Note: May also be tested prone or seated

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- Active Range of Motion
 - Flexion
 - Extension
 - ABDuction
 - ADDuction
 - External Rotation
 - Internal Rotation

Knee Evaluation

- Knee Range of Motion:
 - Flexion 145° - 150°
 - Extension 0° - 5°
 - Rotation varies with flexion
 - Full extension, there is minimal rotation
 - At 90° flexion, 45° ER and 30° IR
 - Varus and Valgus motion is normally limited to a few degrees and should be bilaterally symmetrical

- Knee Active Range of Motion
 - Flexion
 - Extension
 - External Rotation
 - Internal Rotation

Knee Evaluation – Range of Motion

Knee Movement	Primary Muscles Affecting Movement	Patient Instructions
Flexion	Hamstring group: biceps femoris, semitendinosus, and semimembranosus	<i>“Bend or flex your knee.” OR “Squat down to the floor.”</i>
Extension	Quadriceps: rectus femoris, vastus medialis, lateralis, and intermedius	<i>“Straighten your leg.” OR “After you squat down to the floor, stand up.”</i>
Internal Rotation	Sartorius, gracilis, semitendinosus, semimembranosus	<i>“While sitting, swing your lower leg toward the midline.”</i>
External Rotation	Biceps femoris	<i>“While sitting, swing your lower leg away from the midline.”</i>

- Test muscle strength of the knees
 - Flexion
 - Extension

Special Tests - Hip

- Trendelenburg's Test (Gluteus Medius)
- Stinchfield Test (Intra-articular hip pathology)
- Patrick's (FABERE) Test (Hip joint pathology, S-I dysfunction, or Gluteus Medius pathology)
- Log Roll (Non-specific)
- FADIR (Hip joint pathology)
- Thomas Test (Psoas)
- Erichsen's Test (Sacroiliitis)
- Piriformis Test
- Ober's Test (Iliotibial Band Contracture)
- Hip Drop Test (Assess Lumbar Involvement)

Trendelenburg's Test

- The Patient stands first on one leg on the unaffected side, then the affected side, while observing for a continued pelvis
- *Positive Test:* The pelvis on the opposite side drops
- *Indicates:* Inability of the hip abductors to stabilize the pelvis on the femur indicates a weak or painful gluteus medius on the weight-bearing leg

Source: ICC Guide

Video Credit: NYITCOM Bears and Red Wolves

Trendelenburg's Test

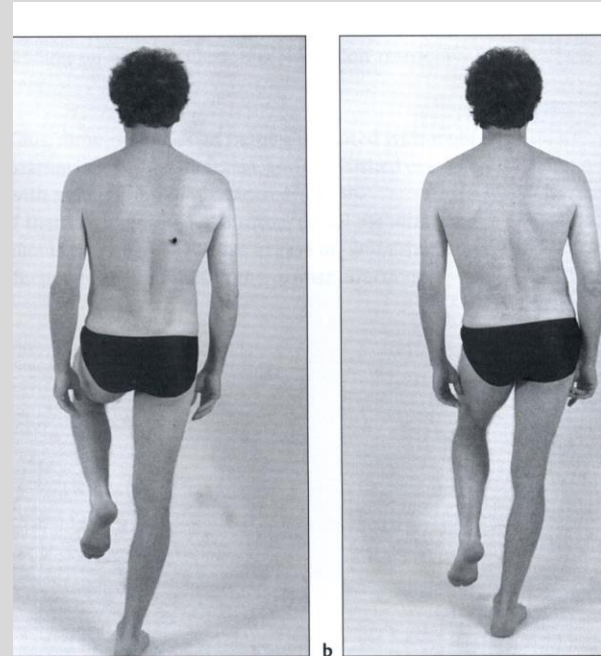
- Abnormal: Iliac crest drops on contralateral side of weakness (gluteus medius weak on standing leg)

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Normal

Abnormal



Stinchfield Test

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- The patient is supine with knee in extension. The patient actively raises leg off table while examiner applies manual resistance.
- Positive test: Reproducible pain usually in the groin, gluteal region and greater trochanter region
- Indicates: Intra-articular hip pathology

Video Credit: NYITCOM Bears
and Red Wolves

Stinchfield Test

- Abnormal:
Reproducible pain usually in the groin, gluteal region and greater trochanteric region



Patrick's (FABER) Test

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- Indicates the motions engaged:
 - Flexion
 - ABduction
 - External Rotation
 - Extension
- The examiner then pushes the knee toward the table to further stress the hip and SI joint. If needed, the 2nd hand can be placed on the opposite ASIS and can press down simultaneously on both sides.
- Positive test: There is pain from these movements or if patient is unable to perform them
- Indicates: Hip joint pathology, SI dysfunction, muscular problem depending on location of the pain

Patrick's (FABER) Test

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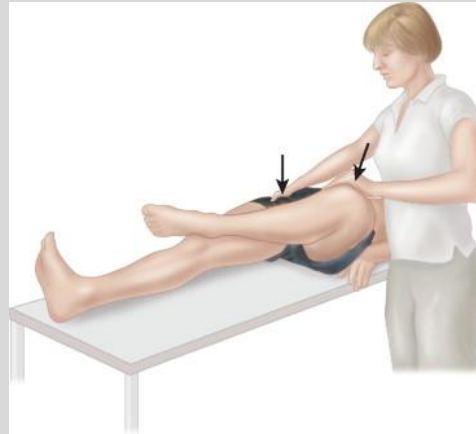
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Patrick's (FABER) Test

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- Also used to stress the sacroiliac joints
- One hand is placed on the abducted knee and the other on the opposite ASIS
- Pain localized to the back when sudden pressure is applied simultaneously on each of these points suggests sacroiliac pathology
- Pain referred to the groin is more suggestive of hip joint pathology



Log Roll

- Nonspecific exam of the hip joint
- Patient supine
- Position the affected leg on the side of the examiner
- The knee is held in extension by the examiner and the foot is internally and externally rotated.
- Positive Test: Pain with motion. A negative evaluation does not preclude hip pathology
- Indicates: Non-specific or more ominously could represent fracture of the hip
- “Popping” sound may be due to acetabular labral tear

FADIR / FAIR Test

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- Indicates motions engaged: Flexion, Adduction, Internal Rotation
- Look for quantity , quality , and symmetry of motion
- Could indicate problem with hip joint (Impingement) or muscular problem (Piriformis)
- Positive test when there is pain from these movements or if patient is unable to perform them

Video Credit: NYITCOM Bears
and Red Wolves

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Thomas Test

- Patient supine.
 - Flexes both knees to chest.
 - Allows one leg to extend.
 - Measure the distance (of extended leg) between popliteal crease to the table
- Positive test: The patient is unable to bring leg down to full extension or if there are compensatory changes in the lumbar spine with full extension
- Indicates: Hip flexor (usually psoas muscle) hypertonicity

Normal

Abnormal

Thomas Test

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Check distance. Should be on table

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Erichsen's Test

- Patient supine
 - Examiner places hands on ilium bilaterally
 - Applies medial pressure towards sacroiliac joints bilaterally
- Positive test: Pain at the SI joints
- Indicates: Sacroiliitis at the side of pain

Ober's Test

- Patient lying on side with the affected side up.
 - The examiner is behind the patient
 - Hip and knees are flexed
 - The upper leg is passively abducted and extended.
 - The limb is then lowered to the table
 - The hip is stabilized to prohibit the patient from rolling backward and allowing the ITB to slip over the greater trochanter
- Positive test: The leg will not fall to the table or there is a delay in drop
- Indicates: Iliotibial Band (ITB) Contracture

Normal

Abnormal

Sources:

1. Stockard, Alan, Osteopathic Clinical Joint Exam; American Osteopathic Academy of Sports Medicine; Chapter 7
2. Note: Slide reproduced from PowerPoint Presented to Class of 2022- *Foundations of Osteopathic Medicine*- OMM Lab #11 – “Osteopathic Evaluation of the Hip” presented November 14th and 15th, 2018

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Ober's Test – ITB Contracture

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Piriformis Test

- Patient supine
 - Flexed hip and knee
 - Active ABduction against examiner resistance
- Positive test: Pain over posterior aspect of the greater trochanter
- Indicates: Piriformis spasm or pathology

Hip Drop Test (Lateral Lumbar Flexion Test)

- Patient standing with feet approx. 4-6 inches apart
- Doctor behind patient, eyes relatively level to the lumbar region.
- Patient is asked to bend one knee, while both feet remain flat on the floor and the other knee is straight.
- Patient allows the compensatory shifts to occur naturally.
- Doctor notes the amount of drop, of the ipsilateral iliac crest, as well as the fluidity of the movement.
- Compare sides. The side of the greatest drop of the iliac crest, is opposite to the side of the greatest lumbar sidebending. [Or, the lumbar convexity, is on the same side, as the side of the greater amount of drop, of the iliac crest during this test].

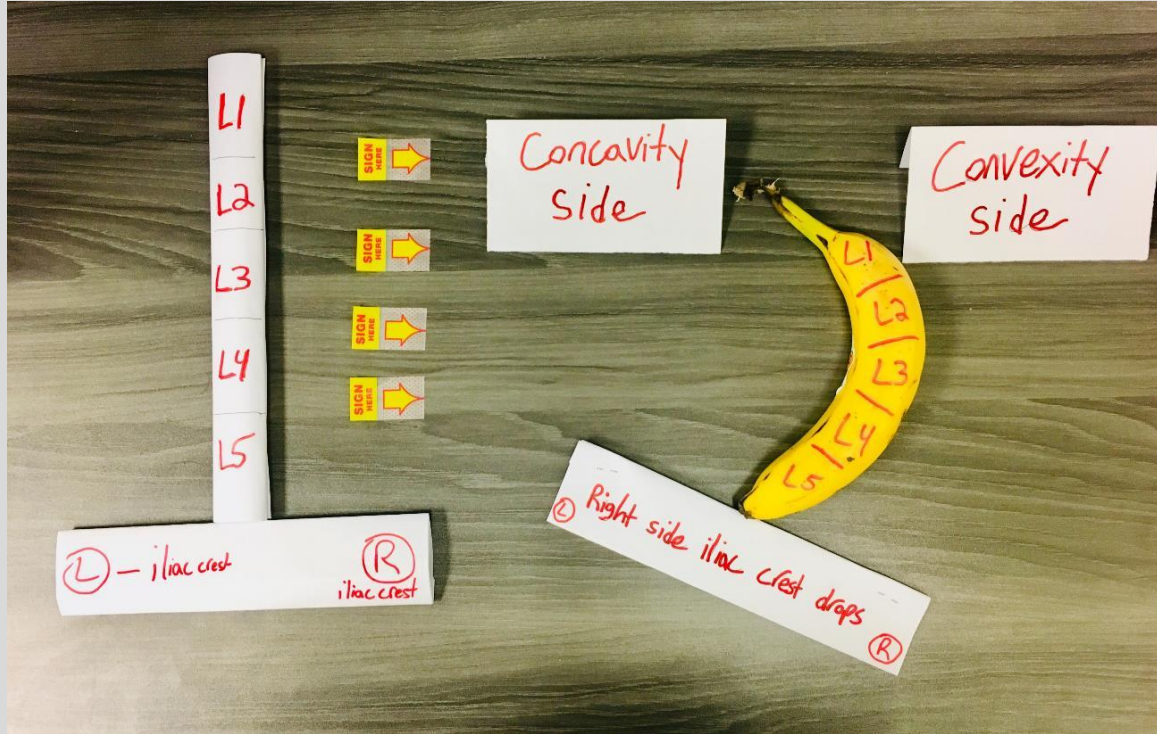
*see DiGiovanna Chapter 47, Evaluation of the Lumbar Spine, p. 239, on library website, search for: “Lateral Lumbar Flexion: Hip Drop Test,” LINK to Ch. 47:

<http://arktos.nyit.edu/login?url=http://meded.lwwhealthlibrary.com.arktos.nyit.edu/content.aspx?sectionid=50019492&bookid=799>

Hip Drop Test (Lateral Lumbar Flexion Test)

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Source: Heller, Matthew. Serota Office 140.

Hip Drop Test (Lateral Lumbar Flexion Test)

- The side of the greatest drop of the iliac crest, is opposite to the side of the side of the greatest lumbar sidebending. [Or, the lumbar convexity, is on the same side, as the side of the greater amount of drop, of the iliac crest during this test].

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Special Tests - Knee

- McMurray's Test (Meniscus)
- Apley Compression Test (Meniscus)
- Apley Distraction Test (Collateral Ligaments)
- Lachman's Test (ACL)
- Anterior Drawer Test (ACL)
- Posterior Drawer Test (PCL)
- Valgus Stress Test (MCL)
- Varus Stress Test (LCL)
- Patellar Laxity Test (Patellar Mechanism)
- Patellar Grind Test (Patellofemoral Joint Disorder)
- Thessaly Test (Meniscus)

McMurray's Test

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- Purpose is to pinch the torn meniscus between the tibial plateau and femoral condyles.
- Examiner lifts the leg of the supine patient, one hand controlling the ankle while the other controls the distal femur
- The knee is flexed
- To stress the **medial meniscus**, the tibia is internally rotated as the knee is extended adding a varus load.
- The examiner is stressing the tibia **medially** so as to compress the **medial** joint compartment, while the rotation forces the condyles on and against the **medial meniscus**
- Positive Test: Pain or a painful click while the knee is brought into extension
- Indicates: Possible meniscus tear (**Medial**)

Video Credit: NYITCOM Bears
and Red Wolves

Source: Stockard, Alan, Osteopathic Clinical
Joint Exam; American Osteopathic Academy
of Sports Medicine; Chapter 8

McMurray's Test

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- Purpose is to pinch the torn meniscus between the tibial plateau and femoral condyles.
- Examiner lifts the leg of the supine patient, one hand controlling the ankle while the other controls the distal femur
- The knee is flexed
- To stress the **lateral meniscus**, the tibia is externally rotated as the knee is extended adding a valgus load.
- The examiner is stressing the tibia **laterally** so as to compress the **lateral** joint compartment, while the rotation forces the condyles on and against the **lateral meniscus**
- Positive Test: Pain or a painful click while the knee is brought into extension
- Indicates: Possible meniscus tear (**Lateral**)

Video Credit: NYITCOM Bears
and Red Wolves

Source: Stockard, Alan, Osteopathic Clinical
Joint Exam; American Osteopathic Academy
of Sports Medicine; Chapter 8

Apley's Compression Test

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- Patient is prone with the knee flexed to 90 degrees
- Standing at the opposite side of the of the involved leg being tested
- The clinician pushes down on the foot to provide a compression force on the meniscus while rotating the foot internally and externally
- Also, can be done with examiner grasping patient's ankle and rotating it internally and externally while a downward force is applied
- Positive Test: Pain during compression plus rotation
- Indicates: Pain on medial or lateral side may indicate meniscal damage on that side

Apley's Distraction Test

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- Patient is prone with the knee flexed to 90 degrees
- Standing at the opposite side of the of the involved leg being tested
- The clinician applies a distraction force pulling up on the foot while rotating the foot internally and externally
- Also, can be done with examiner grasping patient's ankle and rotating it internally and externally while a distraction force is applied
- Positive Test: Pain during distraction force plus rotation
- Indicates: Pain on medial or lateral side may indicate ligamentous injury on that side

Video Credit: NYITCOM Bears
and Red Wolves

Source: Stockard, Alan, Osteopathic Clinical
Joint Exam; American Osteopathic Academy
of Sports Medicine; Chapter 8

Lachman's Test

- Patient is supine
- Standing at the same side of the involved leg being tested
- Bilateral exam
- The Clinician places the proximal stabilizing hand on the distal thigh just above the patella
- The distal mobilizing hand grasps the proximal tibia
- Flex the knee to 15-30 degrees, the examiner uses his distal hand to pull the tibia anteriorly while the proximal hand stabilizes the thigh
- Positive Test: Increased laxity, soft or absent end point
- Indicates: ACL tear / insufficiency

Anterior Drawer Test

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- Standing at the same side of the involved leg being tested
- Bilateral exam
- Patient is supine position with knee flexed to 90 degrees and patient's foot is flat on the table
- The clinician sits on the patient's foot to stabilize the leg while the examiner grasps the proximal tibia with both hands and pulls the tibia anteriorly
- The examiner cups their hands around the knee with the thumbs on the medial and lateral joint lines and the fingers on the medial and lateral insertions of the hamstrings
- Positive Test: Excessive translation when compared to the other knee
- Indicates: ACL tear / insufficiency

Video Credit: NYITCOM Bears
and Red Wolves

Source: Stockard, Alan, Osteopathic Clinical
Joint Exam; American Osteopathic Academy
of Sports Medicine; Chapter 8

Posterior Drawer Test

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- Standing at the same side of the involved leg being tested
- Bilateral exam
- Patient is supine position with knee flexed to 90 degrees and patient's foot is flat on the table
- The clinician sits on the patient's foot to stabilize the leg while the examiner grasps the proximal tibia with both hands and translate the tibia posteriorly
- The examiner cups their hands around the knee with the thumbs on the medial and lateral joint lines and the fingers on the medial and lateral insertions of the hamstrings
- Positive Test: Excessive translation when compared to the other knee
- Indicates: PCL tear / insufficiency

Video Credit: NYITCOM Bears
and Red Wolves

Source: Stockard, Alan, Osteopathic Clinical
Joint Exam; American Osteopathic Academy
of Sports Medicine; Chapter 8

Valgus Stress Test

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- Patient is supine
- Examiner stands on same side of knee being examined
- Bilateral exam
- Examiner slight flexes the patient's knee and moves the thigh to about 30 degrees laterally to the side of the table
- While stabilizing the distal femur with one hand on the lateral side of the knee and the other hand around the medial ankle, push medially against the knee and pull laterally at the ankle to open the knee joint on the medial side
- This creates a valgus load on the MCL of the knee
- Positive Test: Increased laxity, sort or absent endpoint or pain
- Indicates: Medial Collateral Ligament disruption

Video Credit: NYITCOM Bears
and Red Wolves

Source: Stockard, Alan, Osteopathic Clinical
Joint Exam; American Osteopathic Academy
of Sports Medicine; Chapter 8

Varus Stress Test

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- Patient is supine
- Examiner stands at the opposite side of knee being examined
- Bilateral exam
- Examiner slight flexes the patient's knee and moves the thigh to about 30 degrees laterally to the side of the table
- While stabilizing the knee with one hand on the medial side of the knee and the other hand around the lateral ankle, examiner push laterally against the knee and pull medially at the ankle to open the knee joint on the Lateral side
- This creates a varus load on the LCL of the knee
- Positive Test: Increased laxity, sort or absent endpoint or pain
- Indicates: Lateral Collateral Ligament disruption

Video Credit: NYITCOM Bears
and Red Wolves

Source: Stockard, Alan, Osteopathic Clinical
Joint Exam; American Osteopathic Academy
of Sports Medicine; Chapter 8

Patellar Laxity Test / Patellar Apprehension Test

- Examiner positions one hand above and one hand below the joint
- The thumbs are placed against the medial aspect of the patella
- The examiner then pushes the patella laterally, assessing the ROM
- Ask the patient if the maneuver provokes any discomfort or sensation of instability
- Positive test: A positive sense of apprehension or instability
- Indicates: Patellar instability or possible previous patellar dislocation

Normal

Sources:

1. Stockard, Alan, Osteopathic Clinical Joint Exam; American Osteopathic Academy of Sports Medicine; Chapter 8

Patellar Grind test

- Patient is supine with the involved knee extended
- Examiner places the web space of his hand just superior to the patella while applying pressure or can push down on the patella directly
- The patient is instructed to gently and gradually contract the quadriceps muscle
- Positive Test: Pain with movement of the patella or an inability to complete test
- Indicates: Patellofemoral joint disorder

Thessaly Test

- Hold patient's outstretched hands while he or she stands flat-footed on the floor
- Internally and Externally rotated while on 1 leg, with the knee 20 degrees flexed
- Positive Test: Pain with rotational movement
- Indicates: Meniscus Injury

Summary Slide

- Demonstrate the components of the history taking and physical examination of the hip and knee
- Describe subjective and objective findings relating to the examination of the hip and knee
- Distinguish Normal vs. Abnormal examination findings
- Recall the appropriate way to document common normal and abnormal findings of the history and physical Examination

Summary Slide

- Core References:

- Bickley, Lynn S.; Szilagyi, Peter G. *Bates' Guide to Physical Examination and History Taking*, 12e. Wolters Kluwer. 2021. (Chapter 23: The Musculoskeletal System, Section: "The Hip" and "The Knee").
 - <https://nyit.idm.oclc.org/login?url=https://premiumbasicssciences-lwwhealthlibrary-com.nyit.idm.oclc.org/book.aspx?bookid=2964>
- Thompson, Jon C. *Netter's Concise Orthopedic Anatomy*, 2nd Edition. 2010 (Chapter 8: "Thigh/Hip").
- Stockard, Alan, Osteopathic Clinical Joint Exam; American Osteopathic Academy of Sports Medicine; Chapter 7 (Hip) and Chapter 8 (Knee)
- Chamberlain, Rachel, "Hip pain in Adults: Evaluation and Differential Diagnosis." *Am Fam Physician*. 2021; 103(2): 81-89.

Lecture Feedback Form:

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